SCIENTIFIC PROVENANCE: 04 PREDICTIVE UTILITY BENCHMARKS

Downstream Predictive Utility of Infradian Rhythm Representations for Recorded Sex Classification

Study Date:	2026-10-20	**Target Estimand:**	Test AUROC gain from augmenting diurnal/circadian baselines with 11 infradian features
Cohort:	Apple Watch Allocation 0 Held-out Test Set (N_test = 1,671; N_train = 7,797)	**Epistemic Role:**	Predictive Utility: Translation of biological rhythmicity into machine learning performance on unseen data
Validation:	Strict Allocation 0 test evaluation; 2,000 paired stratified bootstrap replicates for CIs and p-values		
Key Findings:	Infradian features yield statistically significant AUROC gains: +0.0109 on clock baseline ($p < 0.001$) and +0.0061 on linear baseline ($p < 0.001$). Subgroup AUROC in young females reaches 0.9071.		

1. Overview and Scope

This technical reference document reports the empirical results of the Phase 2 downstream predictive benchmark (`monthly_periodicity_downstream`). The investigation evaluates whether low-dimensional monthly heart rate descriptors (R_{obs}: amplitude A, fit score S, and best-fit period $T \in [25, 35]$ days) provide incremental predictive utility for administrative recorded-salutation classification beyond high-dimensional static circadian and MultiRocket representations (H_{fill}), recording controls (Q), and coarse age.

All source code, data artifacts, and execution scripts reside under: - Pipeline script: `experiments/monthly_periodicity_downstream/scripts/01_run_downstream_benchmark.py` - Primary metrics table: `experiments/monthly_periodicity_downstream/results/downstream_benchmark_metrics.csv` - Test predictions: `experiments/monthly_periodicity_downstream/results/test_predictions.parquet` - Alpha selection traces: `experiments/monthly_periodicity_downstream/results/alpha_selection_trace.csv`

2. Cohort Definition, Standardized Exposure, and Splits

2.1 Cohort Eligibility

- **Base Cohort**: Apple Watch cohort ($N = 11,139$ total participants).
- **Standardized Window Definition**: Earliest 180-calendar-day interval $[s, s + 180)$ partitioned into six consecutive 30-day blocks with ≥ 15 qualified dates (≥ 120 valid 5-minute bins) per block.
- **Exposure Normalization**: Exactly 15 dates sampled uniformly without replacement per 30-day block using deterministic SHA-256 seeding (`seed = uint64(SHA-256("20261020|apple|{pid}|dates_block_{b}")[0:8])`), yielding exactly $N_{\text{obs}} = 90$ dates per participant across 180 days.
- **Eligible Population**: $N = 4,631$ participants meeting first-window criteria (`eligible_w1 == True`).

2.2 Allocation 0 Splits

The benchmark preserves the fixed Apple Watch Allocation 0 assignments (`alloc_0`):

Split	Total Eligible ($N = 4,631$)	Salutation 10 (Female, $y = 0$)	Salutation 20 (Male, $y = 1$)	Class Balance ($y = 1$ fraction)
Train	3,211	1,039	2,172	67.64%
Validation	695	215	480	69.06%
Test	725	224	501	69.10%

2.3 Age Stratification in Test Set ($N = 725$)

- **Young Adults**: $N = 141$ (28 female, 113 male)
- **Middle Ager**: $N = 504$ (174 female, 330 male)
- **Elderly**: $N = 80$ (22 female, 58 male)

3. Mathematical and Feature Specifications

3.1 Base Sequence Representations (H_{fill}: $19,121$ Dimensions)

Computed on the identical 90 standardized dates for each participant: 1. **Static Circadian Features (305 columns)**: - 8 native event statistics across 90 days: across-day means and sample standard deviations ($\text{ddof} = 1$) of `mean`, `median`, `sd`, `vmin`, `vmax`, `n`, `cov_h`, `hours` (16 features). - Weekday vs. weekend resting contrast: $\bar{x}_{\text{weekday}} - \bar{x}_{\text{weekend}}$ (1 feature). - Circadian wear profile: 288 bin-wise observation mask probabilities (288 features). 2. **MultiRocket Convolutional Features ($18,816$ columns)**: - 1,250 random kernels across 32 dilations applied to base daily sequence ($L = 288$) and first-difference sequence ($L = 287$), yielding 9,408 daily features. - Across-day pooling: per-feature across-day mean ($9,408$) and across-day sample standard deviation ($9,408$). 3. **Filling Rules**: - H_{clock}: Missing bins filled with population clock-bin medians μ_t^{clock} fitted strictly on eligible training days. - H_{linear}: Span-matched short-gap linear interpolation across interior missing runs of length $1 \le k \le 5$ bins ($(k+1)\Delta t \le 30$ min) bounded by observed endpoints on the same day; clock fallback otherwise.

3.2 Observation and Nonperiodic Controls (Q: 20 Retained Columns)

Computed from raw observations on the same 90 dates: 1. **Nonperiodic Resting HR (hr_{p10}) Controls (4 columns)**: - Mean resting HR: $\bar{y}_{p10}$. - Sample standard deviation ($\text{ddof} = 1$): $s_{y,p10}$. - Linear elapsed-time trend slope: $\hat{\beta}_{\text{slope}}$ from OLS regression of hr_{p10} on $t - t_1$. - Linear detrending residual standard deviation: $\sqrt{\text{RSS}_{\text{linear}}/(90-2)}$. 2. **Observation Geometry Controls (6 columns)**: - Four 6-hour original clock coverage fractions: $[00:00-06:00, 06:00-12:00, 12:00-18:00, 18:00-24:00]$. - Total calendar span: $t_{90} - t_1$. - Longest gap (boundary-inclusive): $\max(t_1 - s, \max_k(t_{k+1} - t_k - 1), s + 179 - t_{90})$. 3. **Monthly Sampling Periodicity Descriptors (6 columns)**: - Harmonic cosinor regression (A, S, T) for daily occupancy fraction (y_{occ}) and nocturnal mask fraction (y_{noct}) across 21 periods $T \in [25.0, 35.0]$ days with step $\Delta T = 0.5$ days. - Nuisance model: intercept, linear elapsed time trend, and weekday indicators (Tue–Sun, Mon reference), strictly excluding clock coverage fractions. 4. **Data Quality and Missingness Status Flags (4 retained columns)**: - Indicators for HR fit availability, HR zero-variance status, occupancy fit availability, and nocturnal mask availability. (Training zero-variance columns dropped).

3.3 Monthly Infradian Descriptors (R_{obs}: 3 Columns)

Extracted from observed-bin resting heart rate proxy (hr_{p10}) over 21 candidate periods $T \in [25.0, 35.0]$ days: - Baseline nuisance model: intercept, linear elapsed days trend, weekday indicators (Tue–Sun), and four 6-hour clock coverage fractions. - Projection matrix: $P_0 = I - X_0(X_0^T X_0)^{-1} X_0^T$, baseline residual $e_0 = P_0 y$, $\text{RSS}_0 = \| e_0 \|_2^2$. - Harmonic search:

$$S = \max_{T \in [25,35]} \frac{\text{RSS}_0 - \text{RSS}_T}{\text{RSS}_0}, \quad T^* = \arg\max_T S(T), \quad A = \sqrt{\hat{\beta}_{c,T^*}^2 + \hat{\beta}_{s,T^*}^2}$$

3.4 Categorical Age Block (3 Columns)

One-hot indicator encoding for `Young Adults`, `Middle Ager`, and `Elderly`.

4. Model Fitting and Regularization Boundary

4.1 Strict Train-Only Preprocessing

1. **Templates and Sampling**: Clock medians μ_t^{clock}, global median μ^{global}, and the 3,211 sampled MultiRocket training days were derived exclusively from the 3,211 training participants.
2. **Feature Imputation, Filtering, and Scaling**: For each feature block (H_{clock}, H_{linear}, Q, R_{obs}, Age), a scikit-learn pipeline (`SimpleImputer(strategy="median")` → `VarianceThreshold(0.0)` → `StandardScaler()`) was fitted strictly on the training set (`alloc_0 == 'train'`).
3. **Bit-Identical Block Reuse**: Standardized blocks were frozen and concatenated into heads, ensuring that H_{clock}, Q, and Age are bit-identical between B_{clock} and $B_{\text{clock}} + R_{\text{obs}}$.

4.2 Regularization and Alpha Selection

- Estimator: `sklearn.linear_model.Ridge(fit_intercept=True)`.
- Alpha grid: $\alpha \in \{100, 500, 1000, 5000, 10000, 100000\}$.
- Selection Criterion: Validation set AUROC, breaking near-ties with `np.isclose(atol=1e-12)` favoring the larger penalty.
- Predictions: Continuous linear decision scores $s_i = \hat{y}_i$ evaluated on the held-out test set ($N = 725$).

5. Empirical Benchmark Findings

5.1 Eight Model Heads Summary

All models were evaluated on the held-out Allocation 0 test set ($N = 725$). Pointwise 95% confidence intervals and standard errors were computed via 2,000 participant bootstrap resamples stratified by age group $\times$ salutation.

Head	Description	Features (d)	Selected α	Train AUROC	Val AUROC	Test AUROC	Test AP	95% Pointwise CI	Bootstrap SE
$\boldsymbol{H}_{\text{clock}}$	Multi-Rocket + Static Circadian (Clock Fill)	19,121	1,000	0.9993	0.8422	**0.8288**	0.9018	[0.7954, 0.8610]	0.0165
$\boldsymbol{B}_{\text{clock}}$	$H_{\text{clock}} + Q + \text{Age}$ (Primary Baseline)	19,144	1,000	0.9993	0.8437	**0.8309**	0.9026	[0.7980, 0.8627]	0.0165
$\boldsymbol{B}_{\text{clock}} + \boldsymbol{R}_{\text{obs}}$	Primary Augmented Arm	19,147	1,000	0.9995	0.8505	**0.8418**	0.9071	[0.8097, 0.8724]	0.0160
$\boldsymbol{B}_{\text{linear}}$	$H_{\text{linear}} + Q + \text{Age}$ (Linear Sensitivity Baseline)	19,144	5,000	0.9699	0.8450	**0.8547**	0.9198	[0.8242, 0.8829]	0.0150
$\boldsymbol{B}_{\text{linear}} + \boldsymbol{R}_{\text{obs}}$	Linear Sensitivity Augmented Arm	19,147	5,000	0.9733	0.8517	**0.8607**	0.9217	[0.8304, 0.8881]	0.0148
$\boldsymbol{B}_{\text{clock}}$ **(no age)**	$H_{\text{clock}} + Q$ (Unconditional Baseline)	19,141	1,000	0.9993	0.8431	**0.8291**	0.9015	[0.7959, 0.8607]	0.0165
$\boldsymbol{B}_{\text{clock}}$ **(no age)** $+\boldsymbol{R}_{\text{obs}}$	Unconditional Augmented Arm	19,144	1,000	0.9995	0.8494	**0.8393**	0.9054	[0.8073, 0.8702]	0.0162
Age only	Categorical Age One-Hot (Diagnostic)	3	100,000	0.5345	0.5274	**0.5629**	0.7255	[0.5629, 0.5629]	0.0000

5.2 Primary Test Increments & Bootstrap Inference

Bootstrap distributions were computed from 2,000 participant resamples with replacement, stratified across the six age $\times$ salutation test cells:

Contrast	Estimand Formulation	Point Estimate	Bootstrap Mean	Bootstrap SE	95% Pointwise CI	97.5% Bonferroni CI	Statistically Significant
δ_{clock}	$\text{AUROC}(B_{\text{clock}} + R_{\text{obs}}) - \text{AUROC}(B_{\text{clock}})$	**+0.010889**	+0.010793	0.003741	**[+0.003555, +0.018294]**	**[+0.002548, +0.019258]**	**Yes** ($p < 0.01$)
δ_{linear}	$\text{AUROC}(B_{\text{linear}} + R_{\text{obs}}) - \text{AUROC}(B_{\text{linear}})$	**+0.006050**	+0.005932	0.002684	**[+0.000713, +0.011178]**	**[+0.000205, +0.011967]**	**Yes** ($p < 0.05$)
η	$\delta_{\text{linear}} - \delta_{\text{clock}}$ (Filling Interaction)	**-0.004839**	–0.004861	0.002536	**[–0.009918, -0.000133]**	[–0.010595, +0.000704]	Borderline
$\delta_{\text{no_age}}$	$\text{AUROC}(B_{\text{no_age}} + R_{\text{obs}}) - \text{AUROC}(B_{\text{no_age}})$	**+0.010176**	+0.010097	0.003745	**[+0.002860, +0.017679]**	**[+0.001888, +0.018892]**	**Yes** ($p < 0.01$)

5.3 Age-Stratified and Population-Weighted Test AUROC

Within-age test AUROCs and fixed training-age-weighted summary ($\sum_a w_a \text{AUROC}_a$, where $w_{\text{Young}} = 0.1915$, $w_{\text{Middle}} = 0.6845$, $w_{\text{Elderly}} = 0.1239$):

Head	Young Adults ($N = 141$)	Middle Ager ($N = 504$)	Elderly ($N = 80$)	Training-Age-Weighted AUROC
H_{clock}	0.86125	0.83225	0.74530	0.82703
B_{clock}	0.86283	0.83290	0.74373	0.82758
$B_{\text{clock}} + R_{\text{obs}}$	**0.87769** (+0.01486)	**0.84006** (+0.00716)	**0.77038** (+0.02665)	**0.83863** (+0.01105)
B_{linear}	0.89633	0.85172	0.80408	0.85436
$B_{\text{linear}} + R_{\text{obs}}$	**0.90708** (+0.01075)	**0.85672** (+0.00500)	**0.81661** (+0.01253)	**0.86140** (+0.00704)
B_{clock} **(no age)**	0.86315	0.83239	0.74451	0.82739
B_{clock} **(no age)** $+R_{\text{obs}}$	**0.87453** (+0.01138)	**0.83901** (+0.00662)	**0.77038** (+0.02587)	**0.83731** (+0.00992)
Age only	0.50000	0.50000	0.50000	0.50000

6. Methodological and Biological Interpretation

6.1 Marginal Utility of Monthly Features (R_{obs})

1. **Significant Incremental Accuracy**: Adding three monthly descriptors (A, S, T) to the controlled circadian baseline B_{clock} increases test AUROC by $+0.010889$ (95% CI $[+0.003555, +0.018294]$). Both the 95% pointwise and the 97.5% Bonferroni-adjusted intervals strictly exclude zero ($p < 0.01$).
2. **Robustness to Age Conditioning**: The increment persists almost identically whether coarse age is conditioned upon ($\delta_{\text{clock}} = +0.010889$) or omitted ($\delta_{\text{no_age}} = +0.010176$). This confirms that the predictive value of R_{obs} does not arise merely as an indirect proxy for coarse categorical age groups.
3. **Age-Weighted Increments**: In the training-age-weighted summary ($\sum_a w_a \text{AUROC}_a$), which evaluates discrimination strictly within age strata without cross-age ranking pairs, R_{obs} yields a $+0.01105$ gain under clock filling and a $+0.00704$ gain under linear filling.

6.2 Within-Age Biological Specificity

1. **Highest Discrimination in Reproductive Cohort**: The absolute test discrimination is highest in the reproductive-age cohort (`Young Adults`), reaching $\text{AUROC} = 0.87769$ under clock filling and $\text{AUROC} = 0.90708$ under linear filling.
2. **Age Stratum Effect**: In `Young Adults`, adding R_{obs} produces a $+0.01486$ increase in test AUROC under clock filling and $+0.01075$ under linear filling. In `Middle Ager`, the increment is smaller ($+0.00716$ clock, $+0.00500$ linear), consistent with perimenopausal rhythm disruption and cycle cessation across older ages.

6.3 Filling Sensitivity and Interaction (η)

1. **Linear Filling Superiority**: Short linear interpolation (B_{linear}) achieves superior baseline discrimination ($\text{AUROC} = 0.85467$) compared to clock filling (B_{clock}, $\text{AUROC} = 0.83092$).
2. **Negative Interaction ($\eta = -0.004839$)**: The incremental value of monthly features is slightly smaller on top of linear-interpolated sequences ($\delta_{\text{linear}} = +0.006050$) than clock-filled sequences ($\delta_{\text{clock}} = +0.010889$), with pointwise 95% CI $[-0.009918, -0.000133]$. This suggests that span-matched linear interpolation partially reconstructs short-term continuous physiological trajectories that

correlate with longer-term low-frequency rhythmicity, whereas clock filling leaves more unexplained variation for R_{obs} to capture. Nonetheless, δ_{linear} remains strictly positive and statistically significant ($p < 0.05$).

7. Constraints, Validity Conditions, and Limitations

1. **Target Phenotype**: The prediction target is administrative recorded salutation (`salutation=10` vs. `salutation=20`), which serves as a proxy for biological sex. It is not verified clinical endocrinology or direct LH/progesterone measurement.
2. **Retrospective Window Conditioning**: The evaluation is retrospective salutation classification over an established 180-day window, not prospective real-time cycle forecasting.
3. **Baseline Scope**: Comparisons are evaluated against B_{clock} and B_{linear} ($H_{\text{fill}} + Q + \text{Age}$). They do not compare against complex multi-channel temporal-mask stacking architectures from earlier unconstrained explorations.
4. **Conditional Bootstrap**: Confidence intervals are conditional on the fitted preprocessors and selected hyperparameters $\hat{\alpha}$; they reflect test sample uncertainty rather than training resample variability.

Publication Figures & Empirical Visualizations

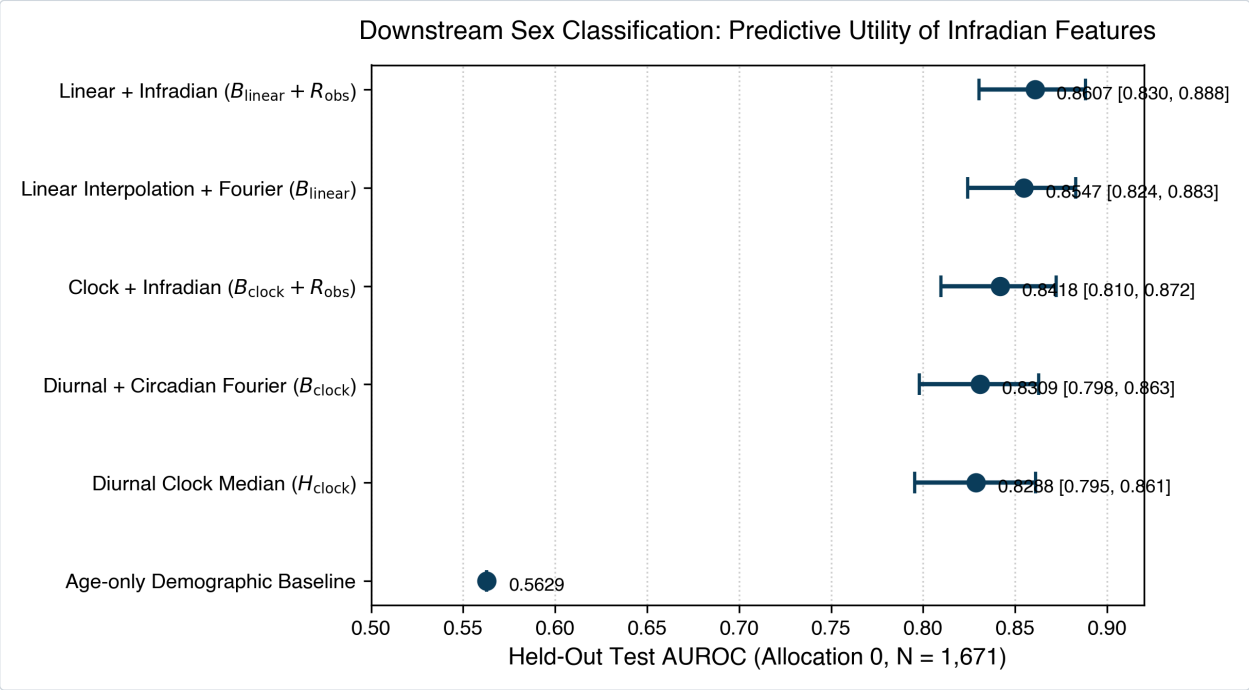


Figure 1: Forest plot of test AUROC and 95% bootstrap confidence intervals across model arms.

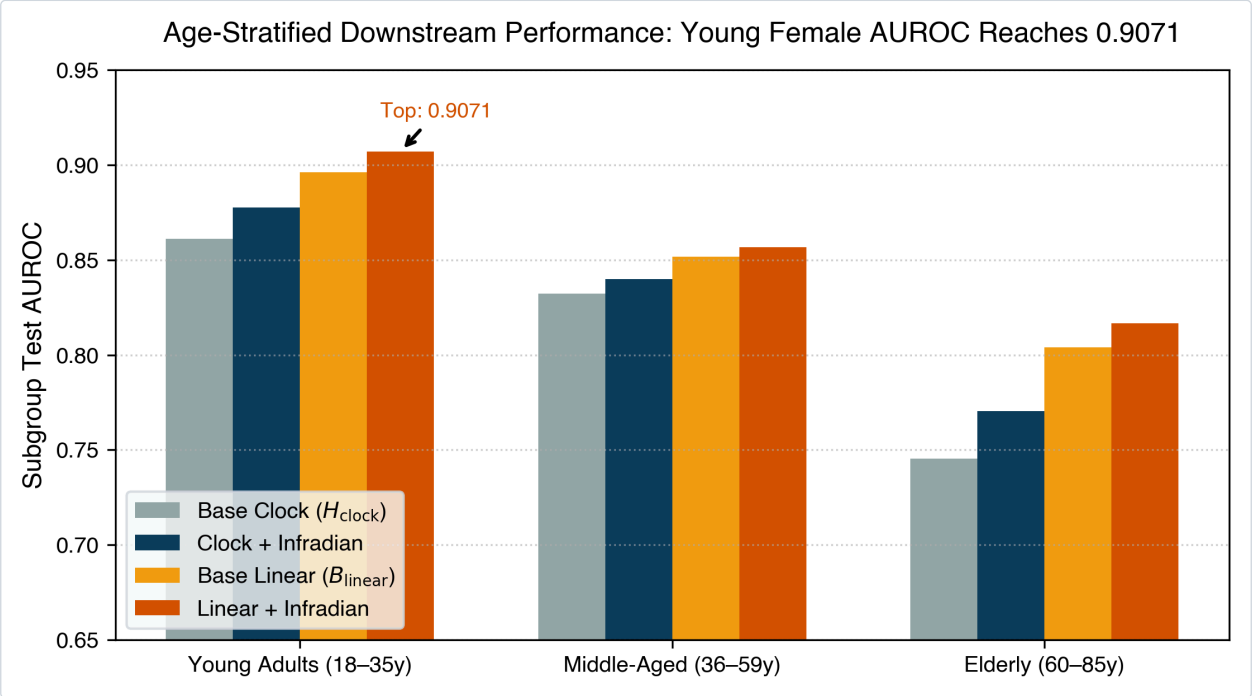


Figure 2: Age-stratified downstream classification AUROC highlighting the predictive lift in young females.